# Gerald (Jace) Curran III

Theoretical condensed matter physicist specializing in spin coherence, triplet exciton dynamics, and Monte Carlo simulation of entangled systems. Published in *PRL* and *Physical Review B.*

## Education

**Doctor of Philosophy in Physics** Ph.D. Defended 2024 – Awarded 2025
*Lehigh University*

Thesis: "The Entangled Triplet Exciton Pair in Rubrene"
Advisor: Ivan Biaggio

**Master of Science in Physics** M.S. Awarded 2022
*Lehigh University*

**Bachelor of Science in Physics (w/ Math minor)** B.S. Awarded 2020
*Kutztown University*

## Publications

**Curran, et al.** ***Persistence of Spin Coherence in a Crystalline Environment*** Published 2024
**Physical Review Letters 133** - https://doi.org/10.1103/PhysRevLett.133.056901
Selected as an Editor's Suggestion

**Curran, et al.** ***Transport-Induced Decoherence of the Entangled Triplet Exciton Pair*** Published 2025
**Physical Review B 112** - https://doi.org/10.1103/rjsk-kjvf

## Research Experience

**Graduate Research Assistant** Summer 2020 – Winter 2025
*Lehigh University*

- Developed theoretical models of transport-induced dephasing (TID) and global decoherence explaining suppression of quantum beats in zero-field and off-axis high-field regimes in crystalline rubrene.
- Devised scheme to use experimentally determined quantum beat frequencies as a probe of zero field parameters (D and E) in acene materials.
- Engineered Monte Carlo Hamiltonian-swapping simulation in MATLAB, to successfully predict observables of fluorescence in acene materials, including quantum beats, TID-driven decoherence, and modulation of fluorescence by an external magnetic field.

**Undergraduate Research Assistant** Spring 2019 – Summer 2020
*Kutztown University*

- Programmed MATLAB simulation to explore the quantum hall effect in simplified conductor models under various topologies.
- Mapped variations in the expected eigenenergy graph (the "Hofstadter Butterfly") to changes in topology, inter-ring coupling, ring radius and perturbations.

## Published Software

**BeatBoostAnalyzer.m** – Companion analysis software for Picoharp .dat time-correlated photon counting files. Can analyze and fit quantum beats, with or without TID decay effects, and can automatically report 'boost' and phase parameters for a given data set with minimal user input.

**SpinSim.m** – Full numerical solution of the triplet pair Hamiltonian given both zero field splitting and electronic Zeeman effect (with small dipole-dipole effect considered). Inputs are unit cell geometry, ZFS parameters, and external field; outputs eigenvalues, singlet projection amplitudes, predicted 'boost' effect, and predicts fluorescence quantum beat frequencies and amplitudes.

## Conference Presentations

**Transport-Induced Dephasing (TID) – Short Talk** APS March Meeting 2025

**Department of Energy PI Meeting – Representative for Biaggio Group** Summer 2025

## Teaching Experience

**Teaching Assistant** September 2020 – January 2025
*Lehigh University*

- Lead lab and recitation sections of 30 students each in introductory physics courses
- Received perfect 5/5 student evaluations, with roughly half of students responding

**Adjunct Professor** Spring 2026
*Moravian University*

## Work Experience

**Database Administrator & Front-End Developer** March 2017 – May 2020
*International Institute for Learning*

- Constructed library of courses and course materials for international teaching organization (more than 500 courses with appendices, handouts, and other materials)
- Rescued at-risk, 1,200 question practice exam project with knowledge of python and xml

**Project Manager / TARP Lead for E.PA** Jan 2014 – Jan 2017
*AT&T*

- Promoted to market TARP lead, and saved AT&T $600,000 through tower review program
- Developed and managed dozens of automated trackers & reports, using VBA for automation